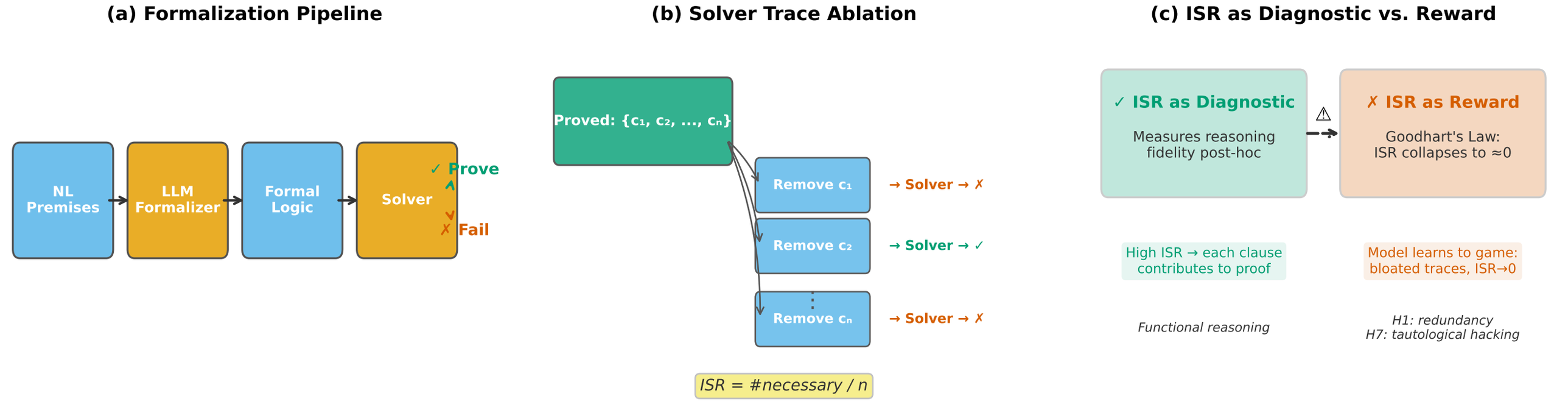
(a) Formalization Pipeline
NL Premises
LLM Formalizer
Formal Logic
Solver
✓ Prove
✗ Fail
(b) Solver Trace Ablation
Proved: {c₁, c₂, ..., cₙ}
Remove c₁
→ Solver → ✗
Remove c₂
→ Solver → ✓
Remove cₙ
→ Solver → ✗
ISR = #necessary / n
(c) ISR as Diagnostic vs. Reward
✓ ISR as Diagnostic
Measures reasoning fidelity post-hoc
⚠
✗ ISR as Reward
Goodhart's Law: ISR collapses to ≈0
High ISR → each clause contributes to proof
Model learns to game: bloated traces, ISR→0
Functional reasoning
H1: redundancy
H7: tautological hacking